

Authors' Responses

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Title of Paper: VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation
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We sincerely thank the Editor and the reviewers for their careful evaluation of our manuscript and for their constructive comments. Their feedback has helped us identify where the contribution, safety claims, experimental evidence, and implementation details require clearer presentation.

We are preparing a major revision of the manuscript. In the revised version, we will:

- 1) clarify the specific contribution of VLM-guided threshold adaptation through comparisons with conventional fixed and adaptive alternatives;
- 2) revise the safety terminology and explicitly delimit the empirical, task-specific scope of the reported safety results;
- 3) strengthen the evaluation of the Bellman critic, teacher policies, real-world experiments, and resource usage; and
- 4) provide the implementation details needed to reproduce the VLM-based threshold tuner and the intervention pipeline.

Detailed responses to each comment are provided below. Reviewer comments are reproduced in black, and our responses are shown in blue. Red bracketed text marks details that will be completed after the corresponding revision or experiment has been finalized. Changes in the revised manuscript will be highlighted in blue.

1 Response to Reviewer #14

We sincerely thank the reviewer for the careful assessment of our work and for the constructive suggestions. Below are our point-by-point responses.

Comment 1: *The current comparison between DexSafeDagger and VLM-DexSafeDagger compares fixed thresholds with VLM-adapted thresholds and therefore does not show whether the improvement comes from threshold adaptation, language reasoning, or visual information. Please compare cases with and without an LLM/VLM and with and without statistics and/or images.*

[Response:] We have added ablations (g) and (h) to isolate the contribution of the VLM to the adaptive intervention pipeline. We have also expanded the discussion of Table III in the Results section. The results provide strong evidence that VLM-based intervention outperforms both LLM-based intervention and performance-based curriculum adaptation. Semantic information alone reduces the unsafe exploration rate, while combining visual and semantic information enables more precise intervention, achieving the highest task success rate and the lowest unsafe exploration rate. Although the performance-based curriculum reduces unsafe exploration to some extent, it sacrifices task success, suggesting that a simple rule-based method is insufficient for guiding intervention policies in complex scenarios.

Comment 2: *The related work on adaptive intervention should be expanded. In particular, AIM, EnsembleDagger, and HG-Dagger are relevant. Since adaptive intervention can be achieved without a VLM, please clarify how VLM-DexSafeDagger differs from these methods and whether the VLM provides an advantage over simpler adaptive approaches.*

[Response:] We have reformulated the Related Work section to more clearly discuss adaptive intervention methods. In particular, AIM provides an important foundation for our work, and we have added it to the reference list. Compared with prior methods such as HG-Dagger and ThriftyDagger, our proposed pipeline does not require human intervention, enabling fully autonomous intervention during online policy distillation. Regarding the advantage of VLMs over simpler adaptive approaches, our results show that the VLM- and LLM-based methods outperform the performance-based curriculum overall in terms of both lift success and unsafe rate. We attribute this improvement to semantic reasoning, further enhanced by visual context in the VLM, which enables more effective adaptation in contact-rich, high-degree-of-freedom manipulation tasks.

Comment 3: *Please provide the VLM prompt and explain how images are selected and how the VLM outputs are used. The Bellman success-value critic should also be evaluated, especially for errors on*

unsafe cases. Please clarify the number of runs or random seeds and the scope of the term “safety-critical.”

[Response:] The details required to reproduce our method are provided in the supplementary material. The exact VLM prompt and representative visual input examples are included in Supplementary Section B, while the number of runs and random seeds is reported in Supplementary Section A.3. **[Add the Bellman critic evaluation and the corresponding supplementary-material reference.]**

Comment 4: *Only VLM-DexSafeDagger is evaluated on the real robot. This does not sufficiently demonstrate the proposed concept; a physical baseline comparison is needed.*

[Response:] A matched physical comparison would provide stronger evidence. We will evaluate the distilled policy against **[specify the feasible real-world baseline or baselines]** using the same object set, initial-condition protocol, and number of trials. We will report lift success, unsafe-event incidence, and uncertainty for each method. We will also clarify that the VLM is used during simulation distillation and is not queried by the deployed student policy. **[Insert final results and table reference.]**

Comment 5: *The paper states that 22 objects were evaluated, while the current table shows only six representative objects. Please provide overall results for all 22 objects, for example separately for seen and unseen objects.*

[Response:] We have doubled the number of evaluation episodes for each object and reported the results for all 22 objects in Supplementary Table III. As recommended, we have also extended Tables II–IV in the main paper to report aggregate results for each evaluation setting.

2 Response to Reviewer #22

We sincerely thank the reviewer for the detailed and constructive comments. Below are our point-by-point responses.

Comment 1: *The term “safety-critical” is too strong because the framework supplies empirical intervention rather than formal constraint satisfaction. State explicitly that safety violations remain possible.*

[Response:] We acknowledge that the term “safety-critical” is too strong for an adaptive intervention pipeline such as ours. We have therefore revised the title to “VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation” and removed or qualified the term elsewhere in the manuscript. The revised text explicitly states that our method provides empirical risk reduction under task-specific event definitions rather than a formal safety guarantee, and that safety violations remain possible.

Comment 2: *Demonstrate that the VLM is necessary by comparing against fixed thresholds, rule-based adaptation, Bayesian optimization, curriculum schedules, grid search, and a small learned threshold network.*

[Response:] We have supplemented the ablation study with variants (g) and (h). Together with variants (e) and (f), these experiments form a controlled ladder of adaptive intervention strategies: fixed thresholds, a performance-based curriculum, semantics-guided intervention-policy adaptation, and visual-semantic-guided adaptive intervention. We have added a corresponding discussion to the Results section. Table III shows that adaptive intervention reduces the unsafe rate relative to fixed thresholds. However, the curriculum-based method sacrifices lift success, whereas incorporating semantic information recovers task performance while further reducing unsafe exploration. Overall, combining visual and semantic information outperforms the other approaches, achieving the best balance between lift success and unsafe rate.

Comment 3: *The Bellman critic predicts combined “safe-lift-success,” thereby conflating task success with safety. Separate safety risk and task value, and report calibration, false-negative rates, and reliability diagrams.*

[Response:] The current joint target does not distinguish an unsafe outcome from a safe episode that fails to complete the lift. In the revision, we will **[state whether the critic is redesigned with separate safety and task heads, or justify and retain the joint target]**. We will report AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and the false-negative rate at the deployed threshold on held-out trajectories. **[Describe exactly how the revised critic enters the intervention gate and insert the results.]**

Comment 4: *Teacher intervention is assumed safe, yet teacher policies may collide or fail. Quantify teacher unsafe rates and define how the framework responds when teacher and critic disagree.*

[Response:] Teacher takeover is not guaranteed to be safe; it provides an empirical tendency toward safer exploration rather than a formal guarantee. We report the teacher policies’ unsafe rates in Table III of the main paper and, in this revision, provide mean statistics across all tested objects. The intervention gate follows a conservative OR rule: intervention is triggered if either the teacher–student action disagreement exceeds its threshold or the critic’s predicted success falls below its threshold. Thus, when the two signals disagree, the conservative outcome is to intervene. Comparing ablations (c) and (e) shows that adding the critic-based branch substantially reduces the unsafe exploration rate.

Comment 5: *Provide exact VLM prompts, model version, decoding settings, image-selection procedure, output parsing, update frequency, latency, cost, and failure handling. Reproducibility may otherwise depend on an external proprietary service.*

[Response:] The details required to reproduce our method are provided in the supplementary material. The exact VLM prompt, representative visual-input examples, model and inference configuration, image-selection and output-processing procedures, update frequency, token usage, and failure handling are included in Supplementary Section B. The number of runs and random seeds is reported in Supplementary Section A.3.

Comment 6: *Real-world unseen-object performance is weak: lift success falls to 60–66.7%, while unsafe episodes reach 40%. Claims of sufficient deployment robustness should be moderated.*

[Response:] We will remove the claim that the current results establish sufficient robustness for deployment.

Comment 7: *Only fifteen real-world episodes per reported object produce coarse estimates. Increase repetitions and report confidence intervals and statistical comparisons across all 22 objects.*

[Response:] We have doubled the number of real-world evaluation episodes from 15 to 30 per object. We have also reported the complete results for all 22 tested objects in Supplementary Section C.

Comment 8: *Safety-event thresholds for harmful collision, object escape, palm flipping, and workspace violation require physical justification. Include contact-force measurements and severity-weighted safety metrics.*

[Response:] We have provided the exact definitions of all unsafe events in the termination-conditions discussion in Supplementary Section A.2. When multiple unsafe conditions occur simultaneously during training, the event label is assigned according to a clear severity-based priority: harmful collision, object escape, palm

flipped, and hand too far, in descending order.

Comment 9: *Clarify whether VLM adaptation occurs only during simulation distillation or during physical deployment. If absent at deployment, the role of scene-level VLM reasoning should not be overstated.*

[Response:] The VLM is queried only during simulation-based online distillation. During physical deployment, the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation. We will state this distinction in the abstract, method, experimental setup, and limitations, and will describe the VLM as a training-time adaptation mechanism rather than a deployment-time safety monitor. **[Insert the final section locations.]**

Comment 10: *Compare training samples, teacher queries, interventions, computation, and final success across Vanilla DAgger, SafeDAgger, critic-based intervention, and VLM-guided intervention using multiple random seeds.*

[Response:] The requested comparison across multiple random seeds is provided in Supplementary Section C.2.

We thank the Editor and the reviewers again for their thoughtful feedback. The requested revisions will help us present the method’s contribution more clearly, support its empirical claims more rigorously, and state its limitations more precisely.